

Test Code Matlab → Python

Initialize Python Environment:

- Importing libraries (numpy, matplotlib)

Define Frequency Range:

- Create frequency array ranging from 0.1[GHz] with 0.01[GHz] steps
- Calculate number of frequency points

Initialize the Layer Properties:

- Define Thickness for each layer
- Define Material for each layer

Create Relative Permittivity and Permeability Arrays:

- Initialize empty arrays for permittivity (epsr) and permeability (mur) with appropriate dimensions
- Fill all constant values for permittivity for layers 1-5 and 9-16
- Add frequency-dependent permittivity for materials 6 to 8 using the complex numbers
- Fill constant values for permeability for layers 1-2 and add frequency-dependent values for materials 3 to 5 and 9 to 16

Handle the Boundary Conditions (Air and Infinite Medium/ air and PEC:

- Set permittivity and permeability for the infinite medium and air/ air and PEC

Prepare Reflection Coefficient Arrays:

- Initialize arrays to store the reflection coefficients (rSlab_TE_abs1, rSlab_TM_abs1) and converted reflection values in dB

Call function RMultiSlab and Iterate over Frequencies:

- Call function RMultiSlab and compute reflection coefficients (rSlab_TE_abs, rSlab_TM_abs) for each layer
- Extract Nvar + 1 reflection coefficient and convert to dB

Plot Results

RMultiSlab Matlab → Python

Import values of number of slabs, relative permittivity for each slab, relative permeability for each slab, frequency, and thickness of each slab:

- Def RMultiSlab

Unit conversions of properties:

- GHz to Hz
- mm to meters

Define Physical Constants:

- Permittivity (ϵ_0) and permeability (μ_0) of free space

Absolute Parameters:

- Compute the absolute permittivity and permeability for each slab

Initialize Arrays:

- Create placeholder arrays for quick processing

Wave Number Calculations:

- Calculate the wave number k for each material based on permittivity and permeability

Initialize Angles:

- Set incidence angle in the first material

Snell's Law:

- Apply Snell's Law iteratively to determine the sine of the transmission angle in each slab

Propagation Constants:

- Compute z-component (k_z) of the wave vector for each slab

Fresnel Coefficients:

- Compute the Fresnel reflection coefficients for both TE and TM polarizations at each interface

Iterative Reflection Calculation:

- Iteratively compute the total reflection coefficients for each slab

Absolute Squared Reflection:

- Compute the squared magnitudes of the total reflection coefficients to obtain physical reflection values

Output: Return reflection coefficients for TE and TM polarizations

